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Laser Ablation-Based Methods for Nanostructuring of Materials[1]

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Abstract—An overview of laser-ablation-based methods for nanofabrication developed by our research group is presented. All methods imply laser-related ablation of material from a solid target and the production of nanoclusters, which are then either deposited on a substrate to form a nanostructured thin film or released to a liquid to form a colloidal nanoparticle solution. Particular attention is given on the absence or presence of the plasma-related absorption of laser radiation, which provides different nano fabrication regimes. The methods lead to the production of nanomaterials, which are of importance for photonics and biosensing applications.

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1. INTRODUCTION

Properties of material are known to be strongly dependent on its chemical nature and the structure of its constituents [1]. In particular, due to the overlapping of atomic or molecular orbitals, bulk materials, composed of a large number of atoms, are characterized by the presence of energy bands, which are responsible for most of physical and chemical properties of solids. However, for nanomaterials with dimension between few to 50 nm, the number of atoms becomes so small that the electronic energy bands are significantly modified, strongly affecting almost all physical properties of the materials (see, e.g., [2–4]). In particular, nanostructured semiconductors such as Si and Ge start to exhibit a strong photoluminescence (PL) emission in the visible [4], while in the bulk state these elements have small and indirect band gaps. Nanostructured metals are also known to exhibit novel interesting properties. In particular, ensembles of metal nanoparticles show spectacular changes of color, as well as transmitted and reflected spectra, when the size, shape of nanoparticles and the average distance between them change [5, 6]. These color effects are attributed to the effective resonant absorption of light by individual metal nanoparticles and the excitation of localized surface plasmons, coupled to collective oscillations of electrons in metals. The employment of new properties of semiconductor or metal-based nanomaterials is expected to lead to significant progress in many areas including optoelectronics, biosensing and imaging, magnetic and optical memory, nanoelectronics, hard coatings, chemical-mechanical polishing, etc.

Lasers offer many attractive possibilities for nanofabrication. When focused on a solid target, intense laser radiation can cause the melting and then the ablation of material from the target, leading to the ejection of atoms and nanoclusters [7]. Further nanocluster growth is determined by both laser parameters (wavelength, pulse length, fluence) and properties of the ambient medium. Here are the following possibilities:

(1) Pure laser ablation when the plasma is transparent to laser radiation. The radiation ablates the target and thus produces nanoclusters, which then cool down through their interaction with ambient environment. If the nanoclusters are produced in a gaseous environment, they can then be deposited on a substrate forming a thin nanostructured film. As another possibility, the ablation in liquids leads to the release of material to the liquid and the formation of a colloidal nanoparticle solution.

(2) The production of laser-assisted breakdown plasma, which strongly absorbs the incoming radiation and reduces the radiation-related ablation efficiency. In general, conditions for the breakdown plasma production exist in both gaseous and liquid environment.

We recently developed a series of laser-assisted nanofabrication methods, which employ both pure laser ablation and laser plasma-assisted phenomena. This paper focuses on the overview of these methods.

[1] The article is published in the original.

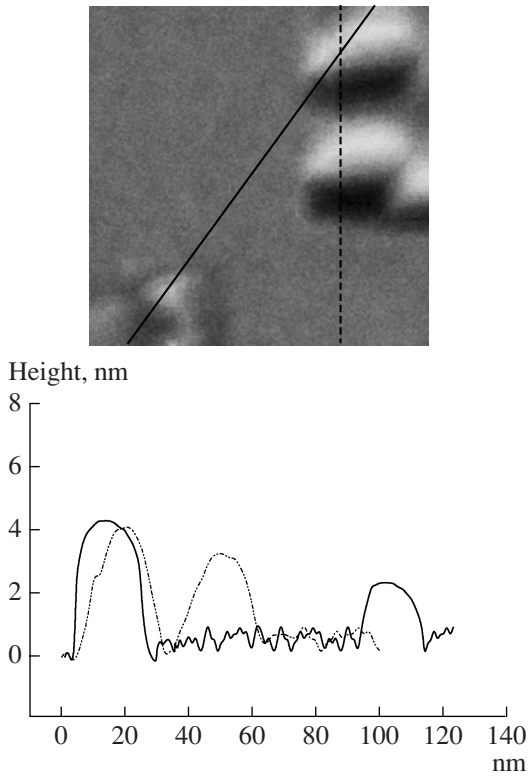

Fig. 1. AFM image of isolated laser-ablated particles on graphite (HOPG) substrate.

2. PURE LASER ABLATION

2.1. Pulsed Laser Deposition (PLD) to Form Nanostructured Films on the Substrate

Historically, most studies on this subject have been devoted to nanostructured silicon films. Such interest is mainly explained by much more promising optical properties of this material compared to bulk silicon. Indeed, both nano-porous silicon [4] and different nanostructured Si-based films (see e.g., [8–11]) exhibit strong visible PL, whereas for bulk Si this PL property is essentially absent. The appearance of PL is usually connected to the presence of nanoscale crystallites in the deposit, while different mechanisms such as quantum confinement [4], surface radiative states [9], defects in SiO_2 structure [12] etc. are considered to explain the visible emission. Taking into account that Si is the leading semiconductor in the modern microelectronics industry, the PL property of nanostructured Si opens up novel opportunities for the creation of a novel generation of inexpensive silicon-based optoelectronics devices.

In our tests, we mainly studied possibilities for the control of properties of deposited films by a variation of radiation parameters [13, 14]. Radiation of a pulsed KrF laser (248 nm, 15 ns FWHM, repetition rate 30 Hz) was used to ablate material from a rotating Si target. The radiation was focused at the incident angle of 45°

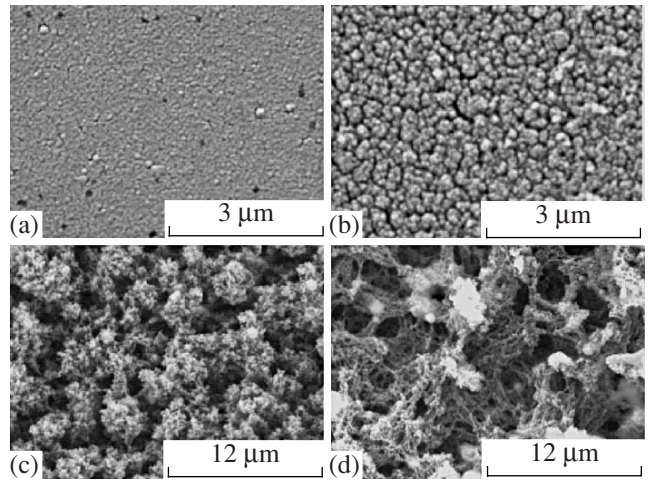

Fig. 2. SEM images of Si-based films fabricated under different pressures of ambient helium: (a) 1, (b) 2, (c) 4, and (d) 8 Torr.

to the surface. The substrate, identical to the target, was placed on a rotating holder in front of the target. The experimental chamber was filled with helium for a deposition at a constant pressure ranging between 0.05 and 10.00 Torr. The film thickness after several thousands laser shots was 100–700 nm.

First, our studies confirmed conclusions of previous works that the size of nanoparticles produced by laser ablation can be reduced down to few nanometers when a residual gas is present during the experiment. A typical AFM image of several isolated laser-ablated particles, deposited on HOPG is shown in Fig. 1, One can see that the height of the particles was about 2–4 nm, which was determined by nanoparticle height measurements. The size of nanoparticles slightly increased under the increase of He pressure.

Secondly, we were first to reveal essentially porous texture of laser-ablation-formed films. We showed that the formation of such texture is affected not only by the size of the ablated particles, but also by conditions of cluster condensation on the substrate. That is, if particles crystallized in the gaseous phase, they arrived on the substrate in the form of a powder. This situation occurred when the pressure of He was too high or the target-substrate distance was relatively large that caused significant cooling of the nanoclusters before reaching the substrate. Basically, the films experienced microfragmentation with the increase of the helium pressure, as one can see from Scanning Electron Microscopy (SEM) images of the laser-ablated films (Figs. 2a–2d). While the deposition under 1 Torr resulted only in some germs of roughness, the experiment under 2 Torr provided a developed porous structure with pore size of about 50–100 nm. Further pressure increase up to 4 Torr led to a formation of web-like aggregations of particles. The porosity of the laser-

ablated films increased with the increase of the helium pressure and exceeded 90% for the films deposited at relatively high pressures.

Another important finding is related to the explanation of the impact of film morphology on PL properties of the films. In particular, we showed that PL signals of the films were quite different for different porosities. Low-porous films $P < 40\%$, deposited at reduced pressures $P < 1.5$ Torr, exhibited relatively weak PL with peak energy strongly depending on the pressure. In particular, the pressure decrease from 1.50 to 0.15 Torr in different depositions caused a blue shift of the peak from 1.60 to 2.15 eV. In contrast, films with an enhanced porosity $P > 40\%$, deposited at higher pressures, provided only spectra with fixed peaks. The first band (1.6–1.7 eV) was independent of the pressure and could be seen just after the fabrication process. In addition, an additional band (2.2–2.3 eV) could appear under the oxidation of samples in humid air. Basically, PL properties of low porous films (porosity <40%) were almost unchangeable under these conditions, while the integral PL intensity from highly porous films significantly increased with the prolonged oxidation. It was concluded that this was the porosity difference that caused the dramatic distinction of long-term PL properties. Dense and self-coagulated structures of the films fabricated under $P < 1$ Torr minimized the impact of ambient atmosphere on the film properties. For these films, mechanisms related to core silicon crystals became predominant, although the upper surface oxide could also play a certain role in a formation of PL centers. In particular, the blue shift of the spectra under the decrease of helium residual pressure was accompanied by a certain decrease of the nanocrystal size, suggesting the involvement of the quantum confinement mechanism [4]. On the other hand, the porosity increased the surface area, which was subjected to surface chemistry modifications due to interactions of nanocrystallites with oxygen and other elements or impurities in ambient air. This could drastically enhance the role of oxidation in the formation of PL centers and a domination of oxygen-related PL mechanisms connected either to defects in the SiO_2 structure (usually, this mechanism provides 2.0–2.4 eV PL [12]) or to the interfacial layer (1.65 eV) [9].

2.2. Ultra-Short Laser Ablation in Liquid Environment to Form Colloidal Nanoparticles

In the case of liquid ambience, laser ablation process leads to the release of nanoclusters to the liquid and the formation of a colloidal nanoparticle solution [7]. In contrast to conventional chemical reduction methods, this method enables to avoid the use of toxic chemical reduction agents to control the growth of the nanoparticles. In particular, the laser ablation-based synthesis can be implemented in pure deionized water. The independence of laser-based synthesis of dirty colloidal chemistry makes it unique for the fabrication of markers of bioanalytes for sensing and imaging applications. However, in pure water or any other aqueous solution exempted of additional chemically active components, the size of nanoparticles produced tends to be relatively large, since a natural coalescence of hot ablated nanoclusters can not be easily overcome. In particular, nanosecond laser-based ablation used in most works, generally gives relatively large (10–300 nm) and strongly dispersed (50–300 nm) particles [15–19]. Although certain size control can be achieved by decreasing the wavelength of pumping radiation [15, 17] or varying the laser fluence [15], the range of size variations stays rather broad in the nanosecond radiation case. Mafune [15] showed that size characteristics of nanoparticles can be improved by adding some surfactants such as sodium dodecyl sulfate (SDS) or CTAB during the ablation. SDS covered the nanoclusters just after their production and thus protected them from further coalescence. However, bioimaging applications of so produced nanoparticles were hardly possible since the surfactants are not biocompatible.

We recently showed that a drastic improvement of the size control of laser-synthesized nanoparticles can be obtained using femtosecond lasers [19, 20]. The experiments were carried out with a Ti:Sapphire laser (110 fs FWHM, 800 nm, 1 μJ/pulse, 1 kHz). The radiation was focused onto Au or Ag rod, which was placed on the bottom of a glass vessel filled with aqueous solutions. The vessel was placed on a horizontal moving platform to avoid the ablation of material from the same place. The ablation experiments were carried out in pure deionized water and in aqueous solutions of different materials: cyclodextrins, biopolymers, salts etc.

First, we found that in contrast to nanosecond ablation regime the size of nanoparticles can be efficiently tuned by simply changing the intensity of fs radiation on the target surface [19, 20]. In particular, the mean particle size could be reduced from 120 to 4 nm when the laser fluence was finely decreased down to ablation threshold values [19]. Here, two different populations of nanoparticles (low and highly size-dispersed ones) were revealed, as shown in Fig. 3, whose appearance was attributed to photon- and plasma-induced ablation mechanisms, respectively (details on the analysis of nanoparticle size distributions in deionized water, and mechanisms of material ablation, can be found in [19]).

As we found from XPS and FTIR studies [21–23], the particles produced were partially oxidized. Furthermore, the oxidized surface of gold nanoparticles can have different termination, depending on the pH of the environment. Under our experimental conditions, the oxidized portion of the gold surface almost exclusively had Au–O groups at pH > 5.8 and increasing numbers of Au–OH groups at pH < 5.8. Being quite different compared to conventional colloidal nanoparticles prepared by chemical reduction methods, such surface chemistry opens up novel opportunities for chemical manipulations. To study this chemistry, we performed

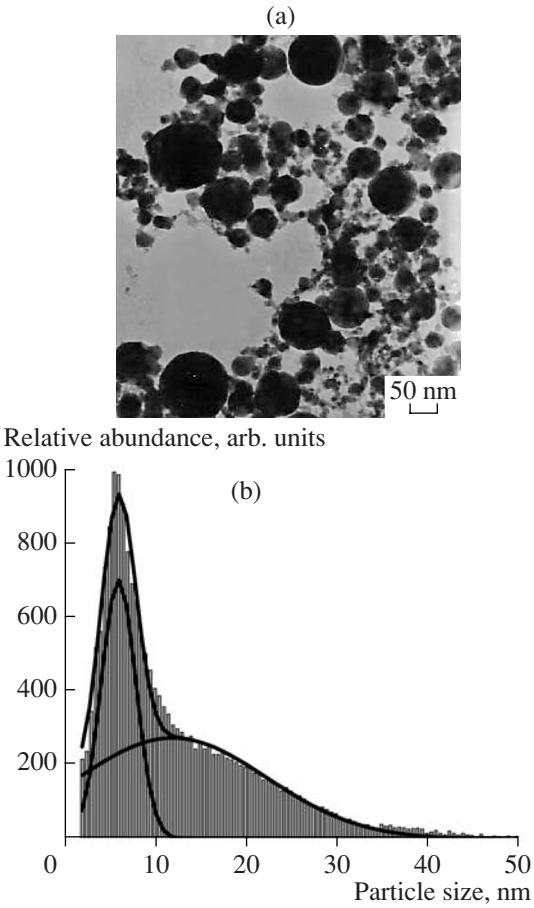

Fig. 3. TEM micrograph image and corresponding size distribution of silver nanoparticles prepared by the femtosecond laser ablation in pure deionized water.

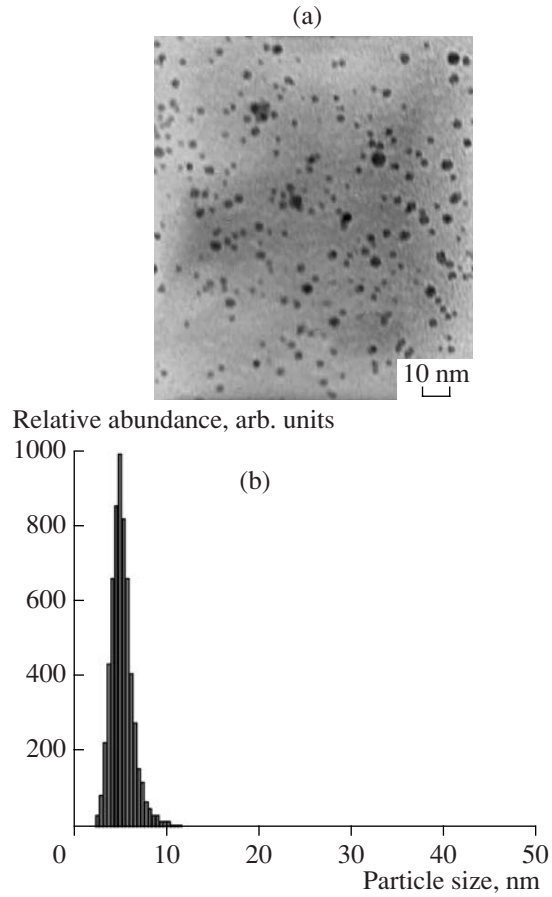

Fig. 4. TEM micrograph images and corresponding size distributions of gold particles prepared by the fs laser ablation in 0.01 M β-CD.

ablation experiments in the presence of chemical substances having different reactive groups.

Thiol group. We performed laser ablation experiments in the presence of 3-mercaptopropionic acid (3-MPA, >99%, Sigma-Aldrich). However, 3-MPA hardly affected the process and was unable to stop the formation of large particles, associated with the second hardly dispersed population of nanoparticles (Fig. 3). It was a surprise since the thiol group is known as the best chemical reagent to stabilize the gold colloids. We believe that such a discrepancy of results for laser-ablated nanoparticles compared to chemically synthesized ones can be explained by a partial oxidation of their surface, which reduced the efficiency of Au-thiol interaction.

OH-group. To understand the influence of OH group, we carried our laser ablation experiments in the presence of oligo-saccharides and various biopolymers (Dextran, polyethylene glycol (PEG)) [20–24]. In particular, very impressive results were obtained using cyclodextrins (α-cyclodextrin (CD), β-CD, and γ-CD. CDs are torus-like macrocycles built up from D–(+)-glucopyranose units (the most common ones are α-CD, β-CD, and γ-CD, consisting of six, seven and eight units, respectively) [25]. Their interior cavity, as well as the primary face, is hydrophobic, while the secondary face and exterior surface are hydrophilic. In our experiments, a drastic size reduction was observed when the concentration of CDs increased, with the smallest size and narrowest dispersion for β-CD, followed by γ-CD, and α-CD. Figure 4 illustrates the effect of the reduction of nanoparticle mean size under the use of β-CD. One can see that the nanoparticles mean size and size dispersion could be reduced down to 2.1–2.3 nm 1 nm FWHM, respectively. The colloidal solutions were extremely stable, while no traces of glucose, a major degradable product of CDs was present in the solution after the ablation process, suggesting that CD molecules remained intact during the experiment. We postulated that the reduction of nanoparticle size during laser ablation in aqueous solutions of CDs is first of all the result of the hydrogen bonding of the –OH groups present on the same face of the CDs and the $–O^–$ at the gold surface and the hydrophobic interaction between the primary face of the CD molecule and the unoxidized gold nanoparticle surface. The CD molecules

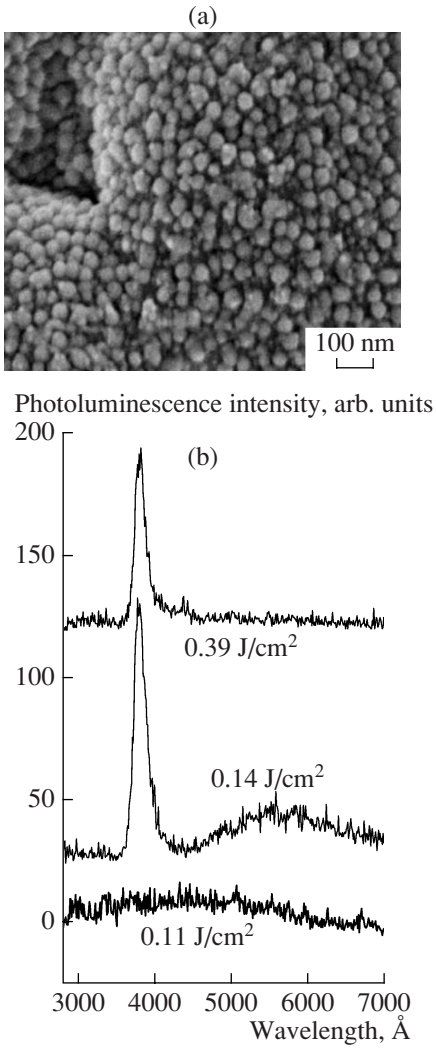

Fig. 5. (a) SEM image of a Zn target after the multi-pulse breakdown productions; (b) Typical photoluminescence spectra from the ZnO layers produced by laser-induced breakdown processing of Zn in air at different fluences.

cover gold nanoclusters just after ablation and act like "bumpers," limiting contact between particles, preventing their coalescence (when the particles are still "hot") and aggregation (when the particles are "cold").

Similar effect of the reduction of nanoparticle size down to 3–7 nm was observed when biopolymers (dextrans, PEG) were used as reducing agents [24]. In this case, the size reduction and stabilisation of solutions mainly comes from the hydrogen bounding between the partially oxidized gold surface and the OH group of the stabilizing agents. Furthermore, binding interactions of different OH groups of dextrans with Au surface can lead to the reinforcement of the resulting effect.

Thus, OH groups of different biocompatible compounds (polysaccharides, biopolymers) can efficiently react with oxidized gold surface leading to the reduction of the nanoparticle size. It is important that in contrast to SDS or other surfactants previously used to control the nanoparticle size [15], CDs and biopolymers are essentially biocompatible. Furthermore, the ultrapure laser-ablated nanoparticles can be functionalized by a proper chemical modification of chemicals. We believe that this gives a huge advantage over the chemically produced nanoparticles for nano-engineering and functionalization of nanoparticles produced, as well as for a solution of toxicity problems.

3. LASER PLASMA-ASSISTED NANOFABRICATION

In most laser nanofabrication methods, one generally uses either ultraviolet or ultrashort radiation to maximize the absorption of radiation by the target itself and avoid the absorption by plasma. In this case, the material experiences a rapid cooling under the interaction with ambient gaseous environment and is deposited to form nanostructured film or powder, while the formed plasma remains almost transparent to the incoming radiation. The result of the ablation process is clearly illustrated by a formation of a deep crater, containing microscale spikes, which is encircled by a mountain of redeposited material [26, 27]. We were the first to study the impact of laser plasma effects on properties of structures spontaneously formed nanostructures during laser-matter interaction [29–32]. Basically, we used the phenomenon of laser-induced gas breakdown, which can be ignited by a simple focusing of infrared laser radiation in atmospheric pressure air, while the presence of a target decreases its initiation threshold by 2–3 orders of magnitude [28]. Absorbing main radiation power the inverse Bremsstrahlung mechanism, the plasma accumulates an enormous amount of energy and is supposed to radically change conditions of nanocluster production and growth. Figure 5 illustrates what happens with the surface of a pure Zn (99.99%) plate if a pulsed CO_2 laser-induced breakdown of atmospheric pressure air is initiated on it. One can see that such a treatment leads to the production of separated and densely packed nanoparticles.

Our experiments also showed that such nanostructures exhibit unique optical properties. In particular, in the case of Si and Ge, these nanostructures can exhibit PL in the visible, which is impossible in bulk state [29, 30]. For ZnO such structures exhibit strong emission around 380–385 nm, which is normally attributed to extremely efficient exciton band, in the absence of defect-related bands [31]. An example of such spectrum is shown in Fig. 5b. Furthermore, we showed that starting from certain pumping intensity several narrow lines within the exciton emission band can be observed [32]. The appearance of these lines is generally attributed to random lasing effect associated with simultaneous high amplification and scattering in a highly disordered medium [33, 34].

CONCLUSIONS

Lasers provide a powerful tool to ablate materials and achieve their nanostructuring. In this cycle of works we developed a series of laser ablation-based methods to synthesize nanomaterials in gaseous and liquid ambiences. Exhibiting unique size-dependent properties, the nanomaterials are of importance for optoelectronics, biosensing and bioimaging applications.

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